

Work Stream 1: Augmented Navier-Stokes & Thermodynamic Admissibility

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1 Introduction

Following discussions on [r/FluidMechanics](#), a consensus emerged: while the Lean 4 proof of finite-time blowup is syntactically flawless, it highlights the need for an augmented admissibility condition in 3D Navier-Stokes, analogous to the Lax entropy condition for the 1D inviscid Burgers equation.

2 The Thermodynamic Admissibility Axiom

We propose that physical solutions to the 3D incompressible Navier-Stokes equations must be bounded by a global thermodynamic constraint. The total dissipation must not exceed physical material limits. Thus, we introduce the *Bounded Enstrophy Condition*:

$$\sup_{t \in [0, T)} \int_{\mathbb{R}^3} |\nabla \times u(x, t)|^2 dx \leq \Omega_{\max} \quad (1)$$

For water at 300K, based on molecular mean free path and thermal dissociation limits, $\Omega_{\max} \approx 1.13 \times 10^{13} \text{ s}^{-2}$.

3 Lean 4 Formalization

As shown in our accompanying `ThermodynamicAdmissibility.lean` file, enforcing this axiom trivially satisfies the Beale-Kato-Majda (BKM) regularity condition. If a solution u is thermodynamically admissible, it cannot blow up. The mathematical singularity manufactured by the OpenAI system requires $\Omega(t) \rightarrow \infty$, thus leaving the set of physically admissible solutions prior to the singularity.

4 Acknowledgments

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